Nexivarin 10 mg Tablets — eCTD Submission

Module 2 — CTD Summaries

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
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2.4 Nonclinical Overview

2.4.1 Overview of the Nonclinical Testing Strategy

The nonclinical development program for nexivarin hydrochloride was designed in accordance with ICH M3(R2), ICH S7A/S7B (safety pharmacology), and ICH S2(R1) (genotoxicity). All GLP-compliant studies referenced herein are reported in full in **Module 4** (Sections 4.2.1 through 4.2.3). The nonclinical written and tabulated summaries are provided in Sections 2.6.1 through 2.6.6 of this Module 2.

Refer to Section 4.1 (Module 4 Table of Contents) for a complete listing of all nonclinical studies submitted. The nonclinical testing strategy and study design rationale are further described in the Nonclinical Written Summaries (Sections 2.6.1, 2.6.3, and 2.6.5).

2.4.2 Pharmacology

Primary Pharmacodynamics

Nexivarin demonstrated potent, concentration-dependent inhibition of L-type calcium channels in vascular smooth muscle cells (IC50 = 2.3 nM) and cardiac myocytes (IC50 = 18.7 nM), confirming peripheral selectivity. The complete in vitro patch-clamp study data are reported in **Section 4.2.1.1** (Study NXV-PD-001). Antihypertensive activity was confirmed in the SHR model with ED50 = 0.8 mg/kg/day (oral), maintained over 4 weeks of daily dosing (Study NXV-PD-002, see Section 4.2.1.1).

Cell Type	Channel	IC50 (nM)	Notes
Human coronary artery SMC	L-type Cav1.2	2.3 ± 0.4	Concentration-dependent; vascular selectivity
Rat ventricular myocyte	L-type Cav1.2	18.7 ± 2.1	~8-fold lower potency vs. VSMC
Human coronary artery SMC	T-type Cav3.1	> 10,000	No significant inhibition
Human coronary artery SMC	N-type Cav2.2	> 10,000	No significant inhibition

As shown in Table 2.4-1, nexivarin is a highly selective L-type calcium channel inhibitor. Full in vitro electrophysiology data and concentration-response curves (Figure 4.2.1.1-1) are provided in Section 4.2.1.1 (Module 4). In vivo antihypertensive dose-response data (Figure 4.2.1.1-2) are also provided in that section.

Secondary Pharmacodynamics

Nexivarin showed no significant activity at concentrations up to 10 µM across a panel of 65 receptors, ion channels, and enzymes (Cerep panel), indicating a clean selectivity profile. The full receptor selectivity data are provided in **Section 4.2.1.2** (Module 4).

Refer to Table 1 of Section 4.2.1.2 (Module 4) for the complete receptor binding panel results. Mild inhibition (<30%) was observed at α1A-adrenoceptors at 10 µM only; this is not considered clinically relevant at therapeutic concentrations, as discussed in Section 2.4.2.

Safety Pharmacology

Cardiovascular safety pharmacology studies (ICH S7B) revealed no QTc prolongation up to 100× the therapeutic Cmax in the rabbit Langendorff model. hERG channel inhibition IC50 was >10 µM (safety margin >1000-fold vs. clinical Cmax). No adverse CNS or respiratory effects were observed. Full safety pharmacology study reports are in **Section 4.2.1.3** (Module 4).

As described in Section 4.2.1.3 (Module 4), cardiovascular, CNS, and respiratory safety were assessed per ICH S7A/S7B core battery. The hERG IC50 (>10,000 nM) and the QTc data table are provided in Table 1 of Section 4.2.1.3. No adverse signals were identified.

2.4.3 Pharmacokinetics

Nonclinical PK parameters for rat, dog, and monkey are summarized in Table 2.4-2 below. Full ADME study reports are provided in **Sections 4.2.2.1 through 4.2.2.5** (Module 4).

PK Parameter	Rat	Dog	Monkey
Oral bioavailability (%F)	45%	82%	71%
Tmax (h)	1.5	3.0	2.5
t½ (h)	8.2	18.5	22.0
Vd (L/kg)	12.4	8.6	9.2
Cl (mL/min/kg)	18.6	6.4	5.8
Plasma protein binding	95.2%	97.1%	96.8%
Primary metabolite	NXV-M1 (inactive)	NXV-M1	NXV-M1

Data listed in Table 2.4-2 are extracted from the full nonclinical PK reports in Module 4 (see Section 4.2.2.2 for absorption studies, Section 4.2.2.3 for distribution, Section 4.2.2.4 for metabolism, and Section 4.2.2.5 for excretion). Metabolite profiling data are provided in Section 4.2.2.4; the metabolic pathway diagram is depicted in Figure 4.2.2.4-1.

2.4.4 Toxicology

Single-Dose Toxicity

The oral LD50 was >2000 mg/kg in both rats and mice. Full single-dose toxicity study reports are provided in **Section 4.2.3.1** (Module 4).

Repeat-Dose Toxicity

In 26-week rat studies (NXV-TOX-006), the NOAEL was 30 mg/kg/day. In 39-week dog studies (NXV-TOX-009), the NOAEL was 10 mg/kg/day (safety margin: 3.0-fold over clinical AUC). Full repeat-dose toxicity study reports are provided in **Section 4.2.3.2** (Module 4).

Species	Duration	NOAEL (mg/kg/day)	AUC Safety Margin
Rat (SD)	26 weeks	30	1.2-fold vs. clinical AUC (see Table 1 in Section 4.2.3.2)
Dog (Beagle)	39 weeks	10	3.0-fold vs. clinical AUC (see Table 2 in Section 4.2.3.2)

As summarized in Table 2.4-3, the NOAEL safety margins are adequate. Detailed toxicokinetic (TK) data including AUC dose-proportionality plots are depicted in Figure 4.2.3.2-1 (rat) and Figure 4.2.3.2-2 (dog), provided in Section 4.2.3.2 (Module 4). Histopathological findings at the high dose are described in Section 4.2.3.2 and are pharmacologically predictable.

Genotoxicity

Nexivarin was negative in the Ames test, in vitro chromosome aberration test (CHO cells), and in vivo mouse micronucleus test. Full genotoxicity reports are in **Section 4.2.3.3** (Module 4).

As described in Section 4.2.3.3 (Module 4), the complete ICH S2(R1) genotoxicity battery yielded negative results (refer to Table 1 in Section 4.2.3.3 for the summary of all genotoxicity study results). The compound is concluded to be non-genotoxic.

Carcinogenicity

Two-year dietary carcinogenicity studies in CD-1 mice and Sprague-Dawley rats showed no compound-related neoplastic findings at exposures up to 37-fold (mice) and 52-fold (rats) above clinical AUC. Reports are provided in **Section 4.2.3.4** (Module 4).

Reproductive and Developmental Toxicity

Fertility and early embryonic development studies showed no adverse effects at doses up to 30 mg/kg/day. No teratogenicity was observed; NOAEL for embryo-fetal development = 30 mg/kg/day. Full reproductive toxicity studies are reported in **Section 4.2.3.5** (Module 4).

Refer to Section 4.2.3.5 (Module 4) for full reproductive toxicity data, including fertility study tables (Table 1 in Section 4.2.3.5) and embryo-fetal development summary (Table 2 in Section 4.2.3.5). The NOAEL and safety margin calculations are provided therein.

2.4.5 Ecotoxicology

Aquatic ecotoxicity studies indicate nexivarin poses minimal environmental risk. The acute LC50 in Daphnia magna was 15.8 mg/L and in zebrafish 22.4 mg/L, both well above predicted environmental concentrations (PEC <0.001 mg/L). The Environmental Assessment is provided in Module 1 (Section 1.7).

Refer to Section 1.7 (Module 1) for the complete Environmental Assessment, including the exposure assessment and risk characterization for nexivarin hydrochloride. Ecotoxicology study reports are referenced therein.